EI Fragment Exact-Mass Calculator

Algorithms, Filters & Fragmentation Rules

6 Post-Filters

Branch-and-Bound Enum.

EI Fragmentation Rules

Reference Implementation — Python / Open Source

Processing Pipeline Overview

1	**Parse SDF Input**	Read molecular formula + EI mass spectrum peaks from SDF file
2	**Electron-Mass Correction**	Convert measured m/z to neutral mass target (EI+/EI−/none)
3	**Branch-and-Bound Enumeration**	Generate all valid candidate elemental compositions (heaviest-first pruning)
4	**Filter Pipeline (Filters 1–6)**	Reject chemically implausible candidates (nitrogen rule, DBE, valence…)
5	**Fragmentation Rule Annotation**	Label candidates with known EI pathways — Tier 1 (neutral loss) & Tier 2 (structure)
6	**Ranking**	Sort by filter pass status → mass accuracy → isotope score
7	**Output**	Annotated text report + enriched SDF file with all candidate formulas

Step 1 — Electron-Mass Correction

Ionisation & Correction

$$M + e^-(\text{fast}) \rightarrow M^{+\bullet} + 2e^-$$

Detector measures the ion, not the neutral:

EI+: $m/z = M_{neut}^{rrr} - m_e$

EI−: $m/z = M_{neut}^{rrr} + m_e$

None: $m/z = M_{neut}^{rrr}$

m_e = 0.000548580 Da

Target neutral mass (EI+):

target = nominal_mz + m_e

Ref: IUPAC 2016 atomic masses; Gross (2017)

Example: Caffeine $[M]^{+\bullet}$

Formula: $C_8H_{10}N_4O_2$

Nominal m/z: 194

Target = 194 + 0.000549 = 194.000549 Da

Actual mass = 194.079602 Da

|194.0796 − 194.0006| = 0.079 Da

→ within ±0.5 Da window ✓

Mode	CLI Flag	Correction
EI+ (default)	--electron remove	$- m_e$
EI−	--electron add	$+ m_e$
None	--electron none	0

Step 2 — Branch-and-Bound Formula Enumeration

Goal

Find all elemental compositions $C^aH^bN^c O^d$ … whose ion mass is within ±tolerance of the observed m/z, subject to atom-conservation (fragment ≤ parent counts per element).

Performance vs. Cartesian Product:

Cartesian product: ~3,000 tested

Branch-and-bound: ~12 valid found

Algorithm: _enumerate_pruned()

1. Sort elements heaviest-first (Cl > S > O > N > C > H)
2. Precompute max_tail[i] = max mass from index i onward
3. **Per level: lo = ⌈(target−tol−partial−max_tail[i+1]) / mass[i]⌉**
   hi = ⌊(target+tol−partial) / mass[i]⌋
4. Prune entire subtrees where lo > hi

Tolerance Modes

Mode	Formula	Use Case
Fixed Da	±0.5 Da	Unit-mass EI (quadrupole)
ppm	mz × ppm / 10^6	Orbitrap / QTOF

Pruning Tree (schematic)

```
Cl (heaviest): lo=0, hi=1  [2 branches]
  ├ Cl=0: reachable → explore ✓
  └ Cl=1: mass > target+tol → PRUNE ✗
S: lo=0, hi=2  [3 branches] → recurse
O, N, C … continue until leaf
Leaf: emit valid formula tuple ✓
```

Ref: Land A.H. & Doig A.G. Econometrica 28(3):497–520 (1960) DOI:10.2307/1910129

DBE — Degree of Unsaturation (Rings + Double Bonds)

$$DBE = 1 + \Sigma_i \; count_i \times (valence_i - 2) / 2$$

Valid: $DBE \geq 0$ and $DBE \times 2 \in \mathbb{Z}$ | Half-integer = odd-electron ion ($M^{+}\bullet$) | Integer = even-electron ion

DBE = 0

Saturated

Alkane, no rings

DBE = 1

One ring or C=O

Cyclohexane / ketone

DBE = 4

Benzene ring

C_6H_6 (3 C=C + 1 ring)

DBE = 0.5

Odd-electron

EI radical cation $M^{+}\bullet$

Formula	Compound	DBE Calculation	DBE
$C_8H_{10}N_4O_2$	Caffeine	1 + 8(1) + 10(−½) + 4(½) + 2(−½×2+2×0)	**6.0**
C_6H_6	Benzene	1 + 6(1) + 6(−½)	**4.0**
C_3H_7Cl	Propyl chloride	1 + 3(1) + 7(−½) + 1(−½)	**0.0**

Ref: Pretsch E. et al. Structure Determination of Organic Compounds, 4th ed. (2009) DOI:10.1007/978-3-540-93810-1

Filter 1 — Nitrogen Rule

Rule (McLafferty & Turecek 1993)

For EVEN-electron ions (integer DBE):

odd m/z ↔ odd nitrogen count

even m/z ↔ even nitrogen count

For ODD-electron ions (half-integer DBE):

rule is INVERTED

Implementation:

n_count = N + P atom counts

even_electron = (DBE × 2) is even

if even_e: passed = (mz_odd == n_odd)

if odd_e: passed = (mz_odd != n_odd)

Examples

m/z 194	$C_8H_{10}N_4O_2$ EE, N=4 even	PASS
m/z 149	$C_8H_9O_2$ EE, N=0 even	FAIL
m/z 77	C_6H_5 OE (inverted)	PASS

CLI: --no-nitrogen-rule to disable | P (phosphorus) also counted as nitrogen-equivalent for the parity check.

Ref: McLafferty F.W. & Turecek F. Interpretation of Mass Spectra, 4th ed. (1993) DOI:10.1002/jms.1190080509

Filter 2 — Hydrogen Deficiency (DBE/C Ratio)

Rule: Reject if **DBE / C_count > max_ring_ratio** *(default 0.5, set via --max-ring-ratio)*

Rationale: for typical organic EI fragments, DBE rarely exceeds half the carbon count. A ratio > 0.5 implies an extraordinarily hydrogen-poor structure unlikely to survive as a stable EI fragment.

$C_8H_{10}N_4O_2$	$C_7H_7^+$	C_6H_{12}	C_6H_6
Caffeine	Tropylium m/z 91	Cyclohexane	Benzene m/z 78
DBE=6 C=8 ratio=0.75	DBE=5 C=7 ratio=0.71	DBE=1 C=6 ratio=0.17	DBE=4 C=6 ratio=0.67
FAIL*	FAIL*	PASS	FAIL*

** With default ratio=0.5. Highly aromatic fragments need --max-ring-ratio 0.7+. CLI: --no-hd-check to disable.*

Configurable: --max-ring-ratio RATIO (default 0.5) | CLI: --no-hd-check to disable

Ref: Pretsch E. et al. Structure Determination of Organic Compounds, 4th ed. (2009) DOI:10.1007/978-3-540-93810-1

Filter 3 — Lewis & Senior Valence Rules

Rule 1 — Valence Parity

Σ (valence$_i$ × count$_i$) must be EVEN

Every covalent bond uses 2 valence electrons. An odd total means no valid molecular graph can be constructed — the structure cannot exist.

Rule 2 — Connectivity Minimum

Σ (valence$_i$ × count$_i$) ≥ 2 × (atom_count − 1)

A connected graph of N nodes requires at least N−1 edges. Below this minimum, the atoms cannot form a single connected molecule.

Valence Table

H	C	N	O	F	Cl	Br	S	P	Si
1	4	3	2	1	1	1	2	3	4

Examples

CH_4O (methanol)	Σval=10 (even) ≥ 8	PASS
CH_3• (radical)	Σval=7 (odd)	FAIL R1
C_2H_2 (acetylene)	Σval=10 ≥ 4	PASS

CLI: --no-lewis-senior to disable | *Ref: Senior J.K. Am. J. Math. 73(3):663–689 (1951) DOI:10.2307/2372318*

Filter 4 — Isotope Pattern Matching

Score = Σ |theoretical % − observed %| for M+1 and M+2 **Lower = better. Threshold: 30 pp (--isotope-tolerance)**

Key Isotope Contributors

Isotope	Abundance	Peak	Element
^{13}C	1.109%	M+1	per C
^{15}N	0.369%	M+1	per N
^{2}H	0.015%	M+1	per H
^{34}S	4.250%	M+2	per S
^{37}Cl	32.50%	M+2	per Cl
^{81}Br	97.30%	M+2	per Br

Examples

C_6H_5Cl (chlorobenzene, M=112):

Theo M+2 = 32.5% (^{37}Cl)

Obs M+2 = 5% → score = 27.5 pp

→ Borderline PASS (< 30 pp)

$C_8H_{10}N_4O_2$ (caffeine, M=194):

Theo M+1 ≈ 9.4% (8 × 1.1%)

Obs M+1 ≈ 10% → score = 0.6 pp

→ Excellent PASS

CLI: --no-isotope-score to disable | --isotope-tolerance PP | Ref: Gross J.H. Mass Spectrometry: A Textbook, 3rd ed. (2017) DOI:10.1007/978-3-319-54398-7

Filter 5 — Structural / SMILES Ring Constraints

Rule: Reject fragment if **DBE_fragment > parent_ring_count × 2 + 1**

A fragment cannot have more rings than the parent. Ring count from V2000 MOL block: rings = bonds − atoms + 1 (cyclomatic number).

Data Source — MOL Block Parser

parse_mol_block_full() extracts:

- Atom table (element, formal charge)
- Bond table (a1, a2, bond type)
- Adjacency graph
- **ring_count = |bonds| − |atoms| + 1**

If no MOL block → filter skipped

Examples

Cyclohexane (rings=1)	DBE=2 ≤ 3	PASS
Cyclohexane (rings=1)	DBE=3 > 3	FAIL
Cholesterol (rings=4)	DBE=8 ≤ 9	PASS
Cholesterol (rings=4)	DBE=9 > 9	FAIL

CLI: --no-smiles-constraints to disable | Ref: Weininger D. J. Chem. Inf. Comput. Sci. 28(1):31–36 (1988) DOI:10.1021/ci00057a005

Filter 6 — RDKit Chemical Validation (Optional, default OFF)

Purpose: Validate element symbols against the RDKit periodic table. Catches impossible symbols (‘Xx’, ‘R’, ‘D’) that slip through earlier filters due to data entry errors.

Implementation

```
from rdkit import Chem
pt = Chem.GetPeriodicTable()
atomic_num = pt.GetAtomicNumber(symbol)
if atomic_num == 0:  → FAIL
```

Graceful degradation:
If rdkit-pypi not installed, filter is silently skipped (returns True).

Install: `pip install rdkit-pypi`

Examples

$C_8H_{10}N_4O_2$	All known	PASS
C_6H_5Xx	'Xx' unknown	FAIL
C_3H_7D	'D' not std	FAIL
$C_{12}H_{22}O_{11}$	Sucrose — OK	PASS

Default: OFF. Enable with --rdkit. Off by default to avoid requiring an optional heavy dependency on non-chem installs.

Ref: RDKit Open-source cheminformatics. https://www.rdkit.org

Fragmentation Rules — Tier 1: Neutral Losses

Formula-based — works for any record, no 2-D structure needed. For each of 30+ tabulated losses: (1) parent contains all atoms, (2) fragment ion mass within tolerance.

Loss	Mass (Da)	Origin
H_2O	18.010565	Alcohols, carboxylic acids, sugars
CO	27.994915	Carbonyl groups, aromatic C=O
CO_2	43.989829	Carboxylic acids, esters
NH_3	17.026549	Amines, amides
HCl	35.976678	Organochlorines — pesticides
HBr	79.926160	Organobromines
HCN	27.010899	Aromatic N, nitriles
CH_3•	15.023475	Alpha-cleavage, methyl esters
CCl_3•	116.907378	DDT-type chloroalkanes
OPCl	98.936978	Chlorophosphate pesticides
$Si(CH_3)_3$ (TMS)	73.046928	GC-MS silyl derivatives
SO_2	63.961901	Sulfones, sulfoxides

Example

Caffeine

$C_8H_{10}N_4O_2$

M^{+}• = 194 Da

m/z 166:

194.080

− 27.995 (CO)

= 166.085 Da

Matched:

$[M–CO]^{+}$

$C_7H_8N_4O$

CLI: --fragmentation-rules | Ref: McLafferty & Turecek (1993); Gross (2017); Budzikiewicz & Grigsby J. Am. Soc. Mass Spectrom. 17:1161 (2006)

Fragmentation Rules — Tier 2: Structure-Based Cleavages

Requires: parsed V2000 MOL block. Uses BFS connected-component analysis after virtual bond cleavage to enumerate fragment compositions.

1. Homolytic Cleavage (σ-bond)

Break every non-aromatic, non-ring single bond. Both fragments carry one radical electron.

$CH_3{-}CH_2{-}OH \rightarrow CH_3\bullet + \bullet CH_2OH$

2. Alpha (α) Cleavage

For each heteroatom (N, O, S, halogen): break adjacent C–C bond. Charge stays on heteroatom fragment.

$R{-}CO{-}CH_2R' \rightarrow [R{-}C{\equiv}O]^+ + \bullet CH_2R'$ *(acylium)*

3. Inductive (i) Cleavage

β-Bond cleavage directed by heteroatom induction. Charge migrates away from the heteroatom.

$R{-}X \rightarrow R^+ + X\bullet$

4. McLafferty Rearrangement

Requires C=O + α + β + γ-H pattern. H migrates from γ-C to O; α–β bond cleaves → enol ion.

Methyl pentanoate → $[CH_2{=}CHOH]^+$ *(m/z 60) + propene*

CLI: --fragmentation-rules to enable (OFF by default). Ref: McLafferty & Turecek (1993); Gross (2017) Ch. 6; Budzikiewicz & Grigsby (2006)

Candidate Ranking — Best-First Ordering

1

Filter Pass Status

filter_passed = True
ranked before
filter_passed = False

Chemically valid structures always shown first.

2

Mass Accuracy

|delta_mass| = |ion_mass − nominal_mz|

Smallest absolute deviation wins. Closest mass match gets priority.

3

Isotope Score

score = Σ |theo% − obs%|
for M+1 and M+2

Lower score = better isotope pattern agreement.

--best-only Mode

Keep only the top-ranked candidate per peak. Peaks where the best candidate still fails all filters are silently dropped from output.

Output example:

```
C8H10N4O2  194.0796 Da  Δ=+0.079  DBE=6.0  OK  [M−CO]
```

Tolerance Modes — Fixed Da vs. Parts Per Million (ppm)

Fixed Da Mode (default)

tol = constant Da (e.g. ±0.5 Da)

CLI: --tolerance 0.5

Same absolute window at every m/z.
Simple, conservative, forgiving.

Best for:
Unit-mass EI instruments
(quadrupole, ion trap)

ppm Mode (high-resolution)

tol = mz × ppm / 1,000,000

CLI: --ppm 5

Per-peak window scales with m/z.
Tighter and more selective.

Best for:
High-resolution EI instruments
(Orbitrap, QTOF)

m/z	±0.5 Da window	±5 ppm window	ppm is tighter by
50	±0.500 Da	±0.000250 Da	2,000×
100	±0.500 Da	±0.000500 Da	1,000×
200	±0.500 Da	±0.001000 Da	500×
500	±0.500 Da	±0.002500 Da	200×

--ppm and --tolerance are mutually exclusive flags. Default is ±0.5 Da if neither is specified.

Summary — Filters & Algorithms at a Glance

Algorithm / Filter	Active by Default	CLI Flag
Electron-mass correction	Always	--electron none
Branch-and-bound enumeration	Always	—
DBE validity (≥0, ×0.5)	Always	—
Filter 1: Nitrogen Rule	ON	--no-nitrogen-rule
Filter 2: H-Deficiency (DBE/C)	ON	--no-hd-check
Filter 3: Lewis & Senior Valence	ON	--no-lewis-senior
Filter 4: Isotope Pattern Score	ON	--no-isotope-score
Filter 5: SMILES / Ring Constraints	ON	--no-smiles-constraints
Filter 6: RDKit Validation	OFF	--rdkit
Fragmentation Rules (Tier 1+2)	OFF	--fragmentation-rules
ppm tolerance mode	OFF	--ppm N

Key References: McLafferty & Turecek (1993) · Pretsch et al. (2009) DOI:10.1007/978-3-540-93810-1 · Senior (1951) DOI:10.2307/2372318 · Gross (2017) DOI:10.1007/978-3-319-54398-7 · Weininger (1988) DOI:10.1021/ci00057a005 · Budzikiewicz & Grigsby J. Am. Soc. Mass Spectrom. 17:1161 (2006) · Land & Doig (1960) DOI:10.2307/1910129 · RDKit: rdkit.org